COURSE PROJECT: NATURAL LANGUAGE PROCESSING

Semester I – Academic Year 2026–2027

Department of Computer Science
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Current Version: v1.1

Version History

Version	Data	Update Description
1.0	2026-30-08	Initial release of the document

1 Introduction

The course project aims to provide students with hands-on experience in designing and implementing a **Natural Language Processing (NLP) system** for a real-world problem.

Through this project, students will learn how to:

- Collect and analyze textual data
- Design a complete NLP processing pipeline
- Apply machine learning or deep learning techniques
- Evaluate and analyze the performance of NLP models

Students are encouraged to explore practical NLP applications using either **English or Vietnamese datasets**.

2 Objectives

After completing this project, students should be able to:

- Formulate a real-world problem as an NLP task
- Implement a complete NLP pipeline
- Apply machine learning and/or deep learning models for NLP
- Evaluate and analyze experimental results

3 Project Description (Open-ended Project)

Students will build an **NLP system for a practical application**.

Each group may choose **one of the following directions** or propose another NLP task subject.

3.1 Text Classification

Examples include:

- Sentiment analysis for product reviews
- News topic classification
- Spam email detection

3.2 Information Extraction

Examples include:

- Named Entity Recognition (NER)
- Information extraction from resumes
- Extracting structured information from news articles

3.3 Semantic Analysis

Examples include:

- Semantic similarity detection
- Paraphrase detection
- Question answering systems

3.4 NLP Applications

Examples include:

- Chatbot systems
- Automatic text summarization
- Fake news detection

4 Technical Requirements

The system must include the main components of a **typical NLP pipeline**.

4.1 Data Collection

Students must either:

- use an existing dataset, or
- collect their own dataset.

Possible sources include:

- Kaggle
- HuggingFace datasets
- GitHub repositories
- public NLP datasets

The report must describe:

- data source
- dataset size
- example samples

4.2 Text Preprocessing

At least **three preprocessing techniques** should be implemented, such as:

- Tokenization
- Stop-word removal
- Stemming or lemmatization
- Word segmentation (for Vietnamese)
- Text normalization

Students should **explain the motivation** for each preprocessing step.

4.3 Text Representation

Students must experiment with **at least two text representation methods**, such as:

- Bag-of-Words
- TF-IDF
- Word2Vec

- FastText
- Sentence-BERT

A comparison between representations should be included.

4.4 Machine Learning or Deep Learning Models

Students must implement **at least two models**.

Possible models include:

Machine Learning

- Naive Bayes
- Logistic Regression
- Support Vector Machines (SVM)

Deep Learning

- Convolutional Neural Networks for text
- LSTM or GRU
- Transformer-based models (e.g., BERT)

4.5 Evaluation

The system should be evaluated using appropriate metrics, such as:

- Accuracy
- Precision
- Recall
- F1-score

Students should include:

- Confusion matrix
- Comparison between models
- Error analysis

5 Deliverables

Each group must submit the following:

5.1 Project Report

The report should include:

1. Introduction

2. Related Work (brief)
3. Dataset Description
4. Methodology
5. Experiments
6. Results
7. Error Analysis
8. Conclusion

5.2 Source Code

The submitted code should include:

- data preprocessing
- model training
- evaluation scripts

5.4 System Demonstration (Optional but encouraged)

Examples include:

- a web demo (e.g., Streamlit)
- an interactive notebook
- a simple API

6 Working Format

Students are required to form groups of 1–5 members. Each group will complete the entire course project (including any extensions, if applicable) according to the registered group.

- **Group registration:** Groups must register their group name and member list in the Google Sheet shared on the course LMS.
- **Supporting evidence:** During the project, groups are encouraged to maintain evidence of collaboration (e.g., images, meeting recordings, discussion notes, etc.) to demonstrate actual teamwork.
- **Task allocation:** In the final report, each group must include a detailed task allocation table for all members, along with their percentage of contribution. Each member's score will be calculated as:

  Individual Score = Group Score x Contribution Percentage (%)

7 Evaluation Criteria

The course project will be assessed based on the following criteria:

Criterion	Weight
Dataset preparation and preprocessing	20%

Criterion	Weight
Model design and implementation	30%
Experiments and evaluation	25%
Report quality	15%
Demo / creativity	10%

8 Submission Deadline & Method

- **Submission method:** The course project must be submitted via the Google Drive folder created and shared by the instructor with the entire class.
- **Deadline:** The specific submission deadline will be announced on the course LMS.